

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – IV

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Class: TYIT	Batch: 1
Date of Assignment: 20-08-2026	Date/Time of Submission: 20-08-2026

Performing Data Analysis using Calculations in Tableau Dataset: Indian Store Data

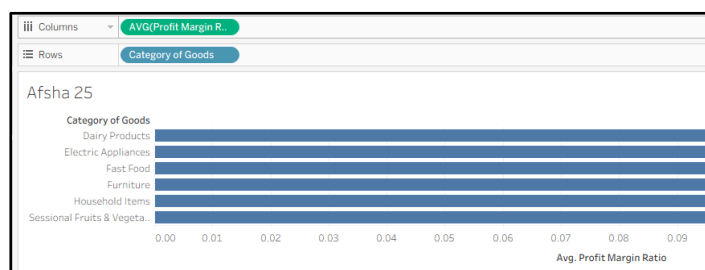
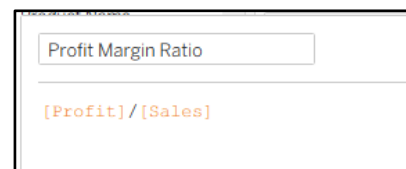
a) Create basic calculated fields using formulas

Step 1: Under the Connect panel, click on Text File, select your store_sales_data.csv, and click Open.

Step 2: Click on Sheet 1 at the bottom-left corner to enter your workspace.

Basic Calculated Field:

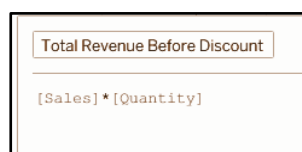
1. Data Pane → Create Calculated Field
2. Name: Profit Margin Ratio
3. Formula: $[\text{Profit}] / [\text{Sales}]$
4. Category of Goods → Rows
5. Profit Margin Ratio → Columns → Average



b) Perform row-level calculations

Row-Level Calculation:

1. Create Calculated Field
2. Name: Total Revenue Before Discount
3. Formula: $[\text{Sales}] * [\text{Quantity}]$
4. Product Name → Rows
5. Total Revenue Before Discount → Columns





c) Perform aggregate-level calculations

Aggregate Calculation:

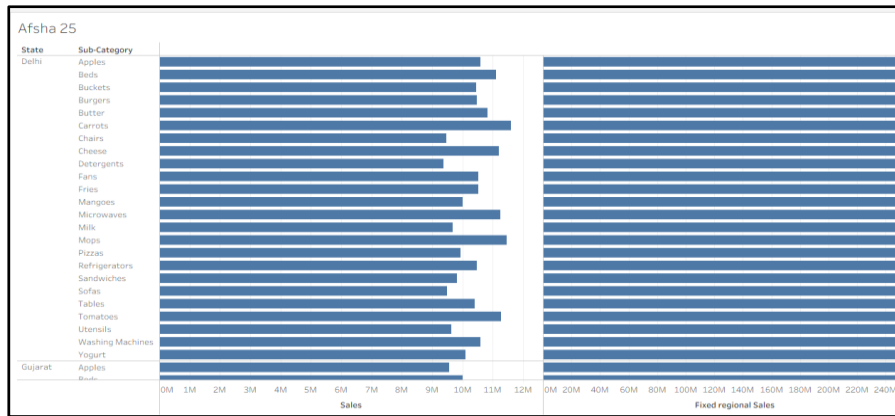
1. Create Calculated Field
2. Name: Aggregate Profit Margin
3. Formula: $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$
4. Sub-Category → Rows
5. Aggregate Profit Margin → Columns



d) Implement FIXED Level of Detail (LOD) calculations

FIXED LOD:

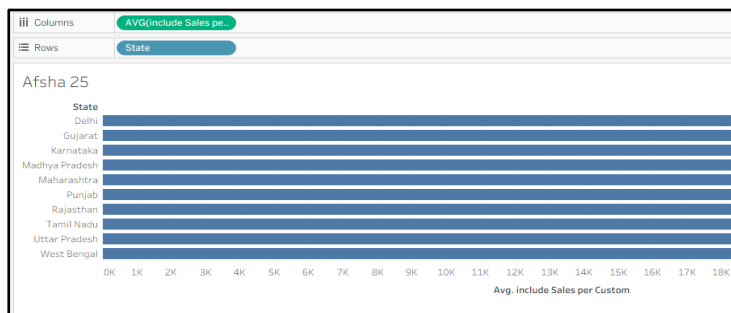
1. Create Calculated Field
2. Name: Fixed Regional Sales
3. Formula: $\{ \text{FIXED } [\text{State}] : \text{SUM}([\text{Sales}]) \}$
4. State + Sub-Category → Rows
5. Fixed Regional Sales → Columns



e) Use INCLUDE and EXCLUDE LOD expressions

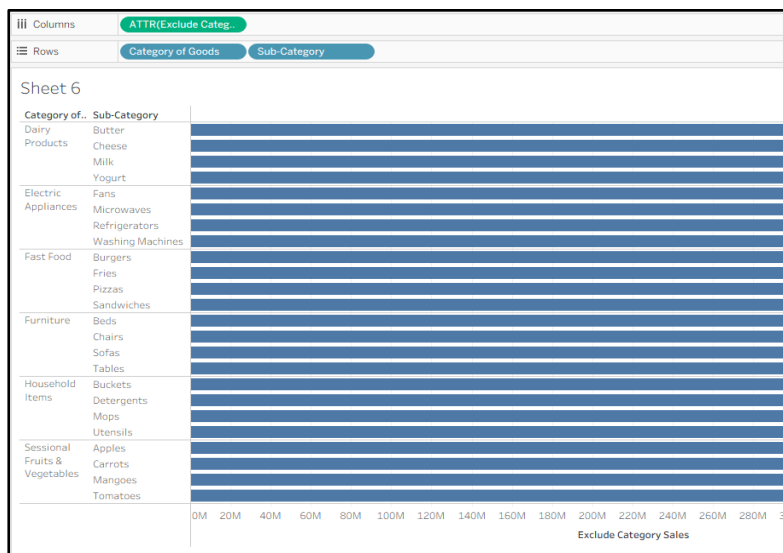
INCLUDE LOD:

1. Create Calculated Field
2. Name: Include Sales per Customer
3. Formula: `{ INCLUDE [Customer ID] : SUM([Sales]) }`
4. State → Rows
5. Include Sales per Customer → Columns → Average



EXCLUDE LOD:

1. Category of Goods + Sub-Category → Rows
2. Create Calculated Field
3. Name: Exclude Category Sales
4. Formula: `{ EXCLUDE [Sub-Category] : SUM([Sales]) }`
5. Exclude Category Sales → Columns

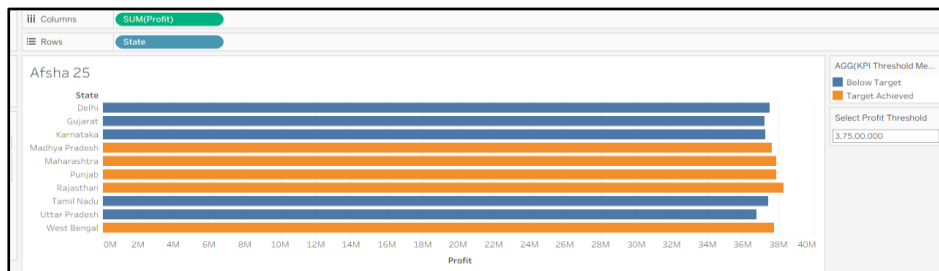


f) Create and use parameters for dynamic analysis

Parameter:

1. Data Pane → Create Parameter
2. Name: Select Profit Threshold
3. Type: Float/Integer, Value: 500
4. Create KPI Threshold Met?
5. Formula:

```
IF SUM([Profit]) >= [Select Profit Threshold] THEN "Target Achieved"
ELSE "Below Target" END
```
6. State → Rows, Profit → Columns
7. KPI Threshold Met? → Color
8. Show Parameter Control



g) Apply calculations to fix data issues and enhance insights

Data Cleaning: Clean Null Values

1. Create **Cleaned Discount**
2. Formula: `IFNULL([Discount], 0)`

Standardize Text

1. Create **Standardized City Type**
2. Formula: `UPPER([City Type])`
3. **Standardized City Type** → Rows
4. **Cleaned Discount** → Columns → Average

